PGR111, tasks for week 10

Problem 1 – concepts & terminology

Briefly explain the following terminology and database terms:

- Determinant and functional dependency (preferably also with an example)
- First normal form (1NF), including the rule
- Second normal form (2NF), including the rule
- Third normal form (3NF), including the rule
- Boyce-Codd normal form (BCNF), including the rule
- Denormalization

Note: Keep your answers. They might become helpful later in the semester. :-)

Task 2 – Normalization to BCNF

Feel free to collaborate with others on the tasks (but you can do the tasks alone as well). Use pen and paper, Lucidchart or something similar for modelling.

Given two tables: CD + Track. A CD can contain multiple tracks, but a track can only belong to one CD. *Important, a couple constraints for this task:* 1) A CD can only contain one artist or a single group. (We cannot have CDs with a mix of several artists/groups). 2) A track can only have a single author.

The current database structure is as follows: *(underscore means PK, asterisk (*) means FK)*
Table CD: cdnum, cdname, artistnum, artistname, releaseyear, releaselocation, price, playtime
Table Track: cdnum, tracknum, time, artistnum, textauthorid, textauthorname.

a) Figure out a primary key for the Track table.
b) Determine all functional dependencies (or determinants, if you like), and normalize the tables to BCNF. Remove unneeded columns, if any. Add new ones if needed. Create an ER model based on your tables.
c) Write down an explanation of why your solution is at 1NF, 2NF, 3NF, and BCNF.
d) Upload your model to the Canvas forum if you would like to get feedback on it.

Task 3 – Database implementation in MySQL

Implement (write SQL for) the database you have modeled in task 2b, above.

Then enter a couple of CDs with corresponding tracks. (You can find data on google or come up with data yourself.)

Task 4 – SQL statements in MySQL

Write a couple of queries against the database. E.g. :

- Extract all tracks from a CD.
- Artist based on a given track.
- Extract relevant info for a track, based on the text author.
- Etc.